

CODES

1. SENSING DATA OF DHT11 & MAX30100

```
#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"

#include "dht11.h"

#include "max30100.h"

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
#define I2C_PORT 0

//DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

esp_err_t i2c_master_init(i2c_port_t i2c_port){
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FRQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        //Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
    }
}
```

```

        if(result.pulse_detected) {
            printf("BEAT\n");
            printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm, result.spO2);
        }
        //Update rate: 100Hz
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        printf("sensing task priority of Max30100 : %d\n",uxTaskPriorityGet(NULL));
        // Update sensor, saving to "result"
        esp_err_t ret = max30100_update(&max30100, &result);
        if (ret != ESP_OK)
        {
            printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
        }
        else
        {
            // Print a message when sensor data is updated
            printf("Sensor data updated successfully\n");

            // Print the pulse detection status
            printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
"No");

            if(result.pulse_detected)
            {
                // Print a message when a pulse is detected
                printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm,
result.spO2);
            }

        }
        // Update rate: 100Hz
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void dht11(void *data)
{
    while(1)

```

```

    {
        printf("sensing task priority of DHT11 : %d\n",uxTaskPriorityGet(NULL));
        printf("Temperature is %d \n", DHT11_read().temperature);
        printf("Humidity is %d\n", DHT11_read().humidity);
        printf("Status code is %d\n", DHT11_read().status);
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

void app_main()
{
    DHT11_init(GPIO_NUM_4);
    //Init I2C_NUM_0
    ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
    //Init sensor at I2C_NUM_0
    ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
        MAX30100_DEFAULT_OPERATING_MODE,
        MAX30100_DEFAULT_SAMPLING_RATE,
        MAX30100_DEFAULT_LED_PULSE_WIDTH,
        MAX30100_DEFAULT_IR_LED_CURRENT,
        MAX30100_DEFAULT_START_RED_LED_CURRENT,
        MAX30100_DEFAULT_MEAN_FILTER_SIZE,
        MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
        true, false ));

    //Start test task
    xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);
    xTaskCreate(dht11, "dht11", 2048, NULL,2,NULL);
    while(1)
    {
        printf("main task : %d \n",uxTaskPriorityGet(NULL));
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

```

Output:

```
ESP-IDF 5.2 CMD - "C:\MAJOR" x + v
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 59
Status code is 0
main task : 1
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 59
Status code is 0
main task : 1
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 59
Status code is 0
main task : 1
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 59
Status code is 0
main task : 1
sensing task priority of Max30100 : 1
```

When tasks are running in two cores:

```
#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"
#include "dht11.h"
#include "max30100.h"
#include "freertos/semphr.h"
#include "freertos/timers.h"
#include "esp_timer.h"

//static SemaphoreHandle_t xSemaphore = NULL;

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
#define I2C_PORT 0

// #define STATS_BUFFER_SIZE 1024
// #define portCONFIGURE_TIMER_FOR_RUN_TIME_STATS() /* Configure a hardware timer if necessary */
// #define portGET_RUN_TIME_COUNTER_VALUE() xTaskGetTickCount()
```

```

//static char stats_buffer[STATS_BUFFER_SIZE];

//DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

// Define global variables to store task execution times
int64_t get_bpm_time = 0;
int64_t dht11_time = 0;

esp_err_t i2c_master_init(i2c_port_t i2c_port)
{
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FREQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        //Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
        if(result.pulse_detected) {
            printf("BEAT\n");
            printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm, result.spO2);
        }
        //Update rate: 100Hz
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

```

```

        printf("sensing task priority of Max30100 : %d\n",uxTaskPriorityGet(NULL));
        printf("sensing core max :%d \n",xPortGetCoreID());
        // if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE)
        //{
            // Update sensor, saving to "result"
            esp_err_t ret = max30100_update(&max30100, &result);
            if (ret != ESP_OK)
            {
                //printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
            }
            else
            {
                // Print a message when sensor data is updated
                // printf("Sensor data updated successfully\n");

                // Print the pulse detection status
                // printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
                "No");

                if(result.pulse_detected)
                {
                    // Print a message when a pulse is detected
                    // printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm,
                    result.spO2);
                }
            }
        // }

        int64_t end_time = esp_timer_get_time(); // End time in microseconds
        get_bpm_time = end_time - start_time; // Calculate processing time for
        get_bpm task

        // Update rate: 100Hz
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void dht11(void *data)
{
    while(1)
    {

        int64_t start_time = esp_timer_get_time(); // Start time in
        microseconds

```

```

        //printf("sensing task priority of DHT11 : %d\n",uxTaskPriorityGet(NULL));
        printf("sensing core DHT11 : %d\n",xPortGetCoreID());
        // printf("Temperature is %d\n", DHT11_read().temperature);
        // printf("Humidity is %d\n", DHT11_read().humidity);
        //printf("Status code is %d\n", DHT11_read().status);

        // Give the semaphore to signal that data is ready
        //xSemaphoreGive(xSemaphore);

        //printf("Semaphore given by dht11 task\n");

        int64_t end_time = esp_timer_get_time(); // End time in microseconds
        dht11_time = end_time - start_time; // Calculate processing time for
dht11 task

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

/*void print_task_runtime_stats(void *params)
{
    while(1) {
        vTaskGetRunTimeStats(stats_buffer);
        printf("Task runtime stats:\n%s\n", stats_buffer);
        vTaskDelay(5000 / portTICK_PERIOD_MS); // Print stats every 5 seconds
    }
}*/

void app_main()
{
    DHT11_init(GPIO_NUM_4);
    //Init I2C_NUM_0
    ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
    //Init sensor at I2C_NUM_0
    ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
        MAX30100_DEFAULT_OPERATING_MODE,
        MAX30100_DEFAULT_SAMPLING_RATE,
        MAX30100_DEFAULT_LED_PULSE_WIDTH,
        MAX30100_DEFAULT_IR_LED_CURRENT,
        MAX30100_DEFAULT_START_RED_LED_CURRENT,
        MAX30100_DEFAULT_MEAN_FILTER_SIZE,
        MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
        true, false ));

    // Create the semaphore
    // xSemaphore = xSemaphoreCreateBinary();
    /* if (xSemaphore == NULL)

```

```

    {
        printf("Failed to create semaphore\n");
        return;
    }*/
    //Start test task
    // xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);
    // xTaskCreate(dht11, "dht11", 2048, NULL,2,NULL);
    //xTaskCreate(print_task_runtime_stats, "Print Task Stats", 2048, NULL, 1,
NULL);

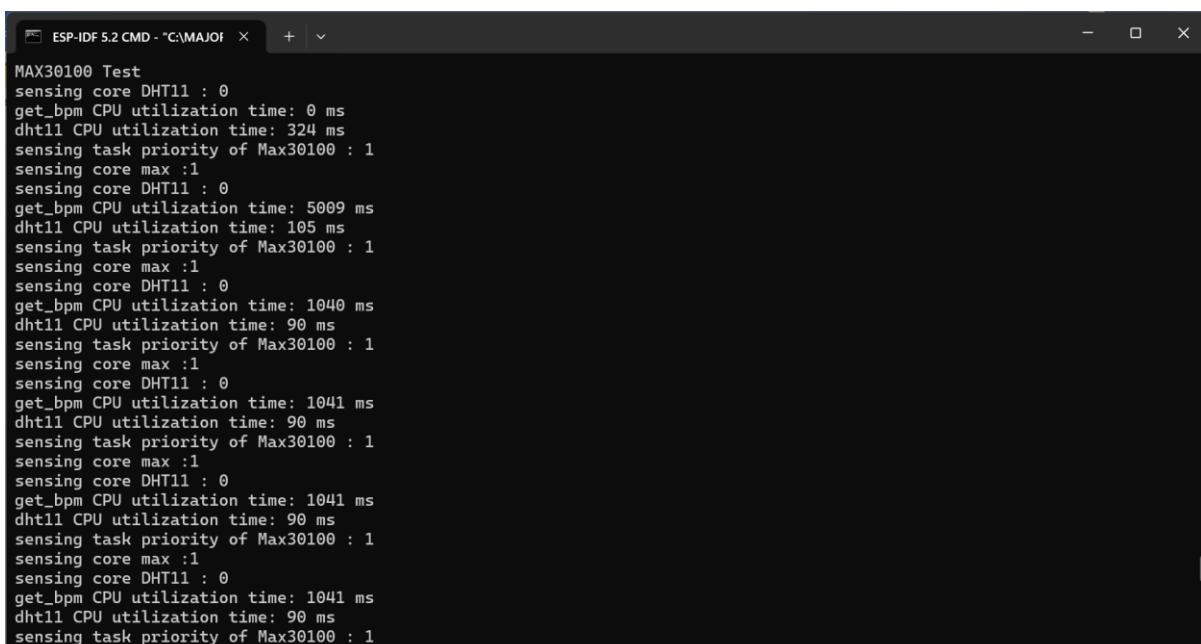
    // Start test tasks and pin them to different cores
    xTaskCreatePinnedToCore(get_bpm, "Get BPM", 2048, NULL, 1,NULL , 1); //
Core 1
    xTaskCreatePinnedToCore(dht11, "dht11", 2048, NULL, 2, NULL, 0);      //
Core 0

    while(1)
    {
        //printf("main task : %d \n",uxTaskPriorityGet(NULL));
        printf("get_bpm CPU utilization time: %lld ms \n", get_bpm_time);
        printf("dht11 CPU utilization time: %lld ms \n", dht11_time);

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

```

Output:



```

ESP-IDF 5.2 CMD - "C:\MAJOF" x + v
MAX30100 Test
sensing core DHT11 : 0
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 324 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 5009 ms
dht11 CPU utilization time: 105 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1040 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1

```

When single core is being used :

```
#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"
#include "dht11.h"
#include "max30100.h"
#include "freertos/semphr.h"
#include "freertos/timers.h"
#include "esp_timer.h"

//static SemaphoreHandle_t xSemaphore = NULL;

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
#define I2C_PORT 0

// #define STATS_BUFFER_SIZE 1024
// #define portCONFIGURE_TIMER_FOR_RUN_TIME_STATS() /* Configure a hardware timer if necessary */
// #define portGET_RUN_TIME_COUNTER_VALUE() xTaskGetTickCount()

//static char stats_buffer[STATS_BUFFER_SIZE];

//DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

// Define global variables to store task execution times
int64_t get_bpm_time = 0;
int64_t dht11_time = 0;

esp_err_t i2c_master_init(i2c_port_t i2c_port)
{
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FRQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}
```

```

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        //Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
        if(result.pulse_detected) {
            printf("BEAT\n");
            printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm, result.spO2);
        }
        //Update rate: 100Hz
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

        printf("sensing task priority of Max30100 : %d
\n",uxTaskPriorityGet(NULL));
        printf("sensing core max :%d \n",xPortGetCoreID());
        // if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE)
        //{
            // Update sensor, saving to "result"
            esp_err_t ret = max30100_update(&max30100, &result);
            if (ret != ESP_OK)
            {
                //printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
            }
        else
        {
            // Print a message when sensor data is updated
            // printf("Sensor data updated successfully\n");

            // Print the pulse detection status
            // printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
"No");

            if(result.pulse_detected)
            {
                // Print a message when a pulse is detected

```

```

        // printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm,
result.spO2);
    }

    }
    // }

    int64_t end_time = esp_timer_get_time(); // End time in microseconds
    get_bpm_time = end_time - start_time; // Calculate processing time for
get_bpm task

    // Update rate: 100Hz
    vTaskDelay(1000 / portTICK_PERIOD_MS);
}
}

void dht11(void *data)
{
    while(1)
    {

        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

        //printf("sensing task priority of DHT11 : %d
\n",uxTaskPriorityGet(NULL));
        printf("sensing core DHT11 : %d \n",xPortGetCoreID());
        // printf("Temperature is %d \n", DHT11_read().temperature);
        // printf("Humidity is %d\n", DHT11_read().humidity);
        //printf("Status code is %d\n", DHT11_read().status);

        // Give the semaphore to signal that data is ready
        //xSemaphoreGive(xSemaphore);

        //printf("Semaphore given by dht11 task\n");

        int64_t end_time = esp_timer_get_time(); // End time in microseconds
        dht11_time = end_time - start_time; // Calculate processing time for
dht11 task

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

/*void print_task_runtime_stats(void *params)
{
    while(1) {

```

```

        vTaskGetRunTimeStats(stats_buffer);
        printf("Task runtime stats:\n%s\n", stats_buffer);
        vTaskDelay(5000 / portTICK_PERIOD_MS); // Print stats every 5 seconds
    }
}*/

void app_main()
{
    DHT11_init(GPIO_NUM_4);
    //Init I2C_NUM_0
    ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
    //Init sensor at I2C_NUM_0
    ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
        MAX30100_DEFAULT_OPERATING_MODE,
        MAX30100_DEFAULT_SAMPLING_RATE,
        MAX30100_DEFAULT_LED_PULSE_WIDTH,
        MAX30100_DEFAULT_IR_LED_CURRENT,
        MAX30100_DEFAULT_START_RED_LED_CURRENT,
        MAX30100_DEFAULT_MEAN_FILTER_SIZE,
        MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
        true, false ));

    // Create the semaphore
    // xSemaphore = xSemaphoreCreateBinary();
    /* if (xSemaphore == NULL)
    {
        printf("Failed to create semaphore\n");
        return;
    }
    */
    //Start test task
    // xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);
    // xTaskCreate(dht11, "dht11", 2048, NULL, 2, NULL);
    //xTaskCreate(print_task_runtime_stats, "Print Task Stats", 2048, NULL, 1,
    NULL);

    // Start test tasks and pin them to different cores
    xTaskCreatePinnedToCore(get_bpm, "Get BPM", 2048, NULL, 1, NULL, 0); //
    Core 1
    xTaskCreatePinnedToCore(dht11, "dht11", 2048, NULL, 2, NULL, 0); //
    Core 0

    while(1)
    {
        //printf("main task : %d \n", uxTaskPriorityGet(NULL));
        printf("get_bpm CPU utilization time: %lld ms \n", get_bpm_time);
        printf("dht11 CPU utilization time: %lld ms \n", dht11_time);

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

```

```
}
```

Output

```
ESP-IDF 5.2 CMD - "C:\MAJOF" x + v
MAX30100 Test
sensing task priority of Max30100 : 1
sensing core max :0
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 236 ms
sensing core DHT11 : 0
get_bpm CPU utilization time: 4125 ms
dht11 CPU utilization time: 94 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3558 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3560 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3560 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3560 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
```

2. CODE AFTER USING SEMAPHORE

```
#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"
#include "dht11.h"
#include "max30100.h"
#include "freertos/semphr.h"

static SemaphoreHandle_t xSemaphore = NULL;

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
```

```

#define I2C_PORT 0

//DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

esp_err_t i2c_master_init(i2c_port_t i2c_port)
{
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FREQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        //Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
        if(result.pulse_detected) {
            printf("BEAT\n");
            printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm, result.spO2);
        }
        //Update rate: 100Hz
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        printf("sensing task priority of Max30100 : %d\n", uxTaskPriorityGet(NULL));
        if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE)
        {
            // Update sensor, saving to "result"
            esp_err_t ret = max30100_update(&max30100, &result);
            if (ret != ESP_OK)
            {

```

```

        printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
    }
    else
    {
        // Print a message when sensor data is updated
        printf("Sensor data updated successfully\n");

        // Print the pulse detection status
        printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
"No");

        if(result.pulse_detected)
        {
            // Print a message when a pulse is detected
            printf("BPM: %f | SpO2: %f%\n", result.heart_bpm,
result.spO2);
        }
    }
}

// Update rate: 100Hz
vTaskDelay(1000 / portTICK_PERIOD_MS);
}
}

void dht11(void *data)
{
    while(1)
    {
        printf("sensing task priority of DHT11 : %d
\n",uxTaskPriorityGet(NULL));
        printf("Temperature is %d \n", DHT11_read().temperature);
        printf("Humidity is %d\n", DHT11_read().humidity);
        printf("Status code is %d\n", DHT11_read().status);

        // Give the semaphore to signal that data is ready
        xSemaphoreGive(xSemaphore);

        printf("Semaphore given by dht11 task\n");
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

void app_main()
{
    DHT11_init(GPIO_NUM_4);

```

```

//Init I2C_NUM_0
ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
//Init sensor at I2C_NUM_0
ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
                                MAX30100_DEFAULT_OPERATING_MODE,
                                MAX30100_DEFAULT_SAMPLING_RATE,
                                MAX30100_DEFAULT_LED_PULSE_WIDTH,
                                MAX30100_DEFAULT_IR_LED_CURRENT,
                                MAX30100_DEFAULT_START_RED_LED_CURRENT,
                                MAX30100_DEFAULT_MEAN_FILTER_SIZE,
                                MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
                                true, false ));

// Create the semaphore
xSemaphore = xSemaphoreCreateBinary();
if (xSemaphore == NULL)
{
    printf("Failed to create semaphore\n");
    return;
}
//Start test task
xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);
xTaskCreate(dht11, "dht11", 2048, NULL, 2, NULL);
while(1)
{
    printf("main task : %d \n", uxTaskPriorityGet(NULL));
    vTaskDelay(1000/portTICK_PERIOD_MS);
}
}

```

OUTPUT :

```
ESP-IDF 5.2 CMD - "C:\MAJOR" x + v
MAX30100 Test
sensing task priority of Max30100 : 1
main task : 1
Temperature is 31
Humidity is 62
Status code is 0
Semaphore given by dht11 task
Sensor data updated successfully
Pulse Detected: No
main task : 1
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 62
Status code is 0
Semaphore given by dht11 task
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
main task : 1
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 62
Status code is 0
Semaphore given by dht11 task
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
main task : 1
```

3. CODE OF CPU UTILIZATION TIME

```
#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"
#include "dht11.h"
#include "max30100.h"
#include "freertos/semphr.h"
#include "freertos/timers.h"
#include "esp_timer.h"

static SemaphoreHandle_t xSemaphore = NULL;

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
#define I2C_PORT 0

// #define STATS_BUFFER_SIZE 1024
// #define portCONFIGURE_TIMER_FOR_RUN_TIME_STATS() /* Configure a hardware timer if necessary */
// #define portGET_RUN_TIME_COUNTER_VALUE() xTaskGetTickCount()

// static char stats_buffer[STATS_BUFFER_SIZE];
```

```

//DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

// Define global variables to store task execution times
int64_t get_bpm_time = 0;
int64_t dht11_time = 0;

esp_err_t i2c_master_init(i2c_port_t i2c_port)
{
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FREQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        //Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
        if(result.pulse_detected) {
            printf("BEAT\n");
            printf("BPM: %f | SpO2: %f%\n", result.heart_bpm, result.spO2);
        }
        //Update rate: 100Hz
        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        int64_t start_time = esp_timer_get_time(); // Start time in
        microseconds

        printf("sensing task priority of Max30100 : %d
\n",uxTaskPriorityGet(NULL));
        // if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE)

```

```

    //{
        // Update sensor, saving to "result"
        esp_err_t ret = max30100_update(&max30100, &result);
        if (ret != ESP_OK)
        {
            printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
        }
        else
        {
            // Print a message when sensor data is updated
            printf("Sensor data updated successfully\n");

            // Print the pulse detection status
            printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
"No");

            if(result.pulse_detected)
            {
                // Print a message when a pulse is detected
                printf("BPM: %f | SpO2: %f%%\n", result.heart_bpm,
result.spO2);
            }
        }
    // }

    int64_t end_time = esp_timer_get_time(); // End time in microseconds
    get_bpm_time = end_time - start_time; // Calculate processing time for
get_bpm task

    // Update rate: 100Hz
    vTaskDelay(1000 / portTICK_PERIOD_MS);
}
}

void dht11(void *data)
{
    while(1)
    {

        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

        printf("sensing task priority of DHT11 : %d
\n",uxTaskPriorityGet(NULL));
        printf("Temperature is %d \n", DHT11_read().temperature);
        printf("Humidity is %d\n", DHT11_read().humidity);
    }
}

```

```

        printf("Status code is %d\n", DHT11_read().status);

        // Give the semaphore to signal that data is ready
        //xSemaphoreGive(xSemaphore);

        //printf("Semaphore given by dht11 task\n");

        int64_t end_time = esp_timer_get_time(); // End time in microseconds
        dht11_time = end_time - start_time; // Calculate processing time for
dht11 task

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

/*void print_task_runtime_stats(void *params)
{
    while(1) {
        vTaskGetRunTimeStats(stats_buffer);
        printf("Task runtime stats:\n%s\n", stats_buffer);
        vTaskDelay(5000 / portTICK_PERIOD_MS); // Print stats every 5 seconds
    }
}*/

void app_main()
{
    DHT11_init(GPIO_NUM_4);
    //Init I2C_NUM_0
    ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
    //Init sensor at I2C_NUM_0
    ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
        MAX30100_DEFAULT_OPERATING_MODE,
        MAX30100_DEFAULT_SAMPLING_RATE,
        MAX30100_DEFAULT_LED_PULSE_WIDTH,
        MAX30100_DEFAULT_IR_LED_CURRENT,
        MAX30100_DEFAULT_START_RED_LED_CURRENT,
        MAX30100_DEFAULT_MEAN_FILTER_SIZE,
        MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
        true, false ));

    // Create the semaphore
    // xSemaphore = xSemaphoreCreateBinary();
    /* if (xSemaphore == NULL)
    {
        printf("Failed to create semaphore\n");
        return;
    }
}*/
//Start test task
xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);

```

```

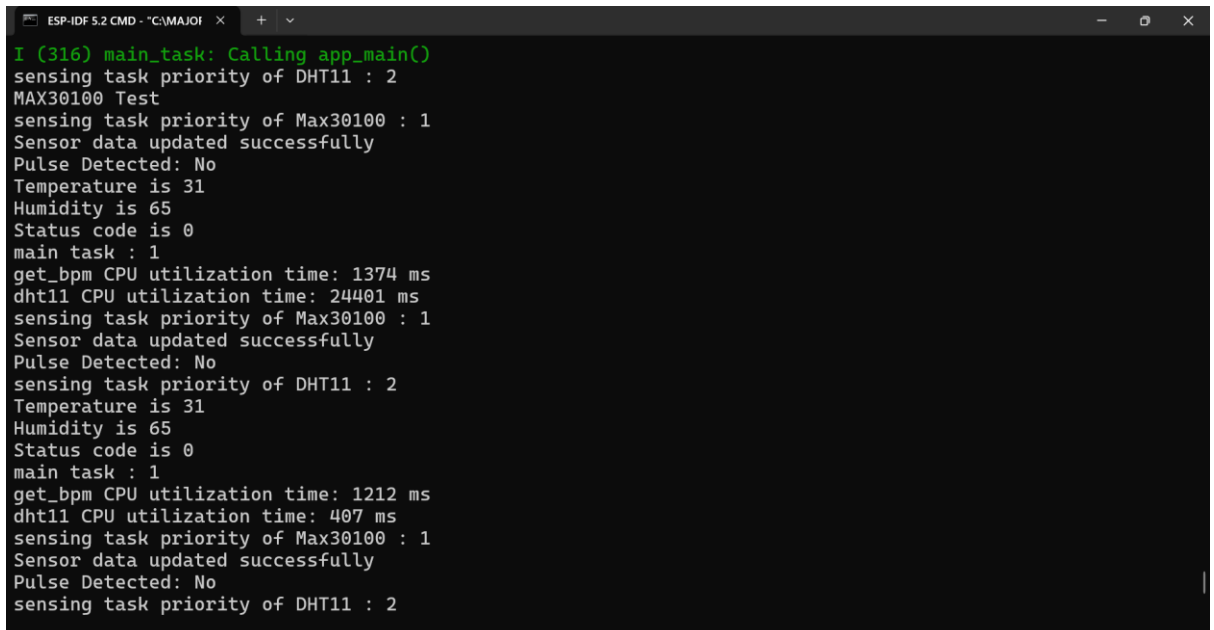
xTaskCreate(dht11, "dht11", 2048, NULL, 2, NULL);
//xTaskCreate(print_task_runtime_stats, "Print Task Stats", 2048, NULL, 1,
NULL);

while(1)
{
    printf("main task : %d \n", uxTaskPriorityGet(NULL));
    printf("get_bpm CPU utilization time: %lld ms \n", get_bpm_time);
    printf("dht11 CPU utilization time: %lld ms \n", dht11_time);

    vTaskDelay(1000/portTICK_PERIOD_MS);
}
}

```

OUTPUT:



```

ESP-IDF 5.2 CMD - "C:\MAJOF" x + v
I (316) main_task: Calling app_main()
sensing task priority of DHT11 : 2
MAX30100 Test
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
Temperature is 31
Humidity is 65
Status code is 0
main task : 1
get_bpm CPU utilization time: 1374 ms
dht11 CPU utilization time: 24401 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 65
Status code is 0
main task : 1
get_bpm CPU utilization time: 1212 ms
dht11 CPU utilization time: 407 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2

```

4. CODE OF CPU UTILIZATION TIME WITH SEMAPHORE

```

#include <stdio.h>
#include "driver/i2c.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "freertos/FreeRTOS.h"
#include "sdkconfig.h"
#include "dht11.h"
#include "max30100.h"

```

```

#include "freertos/semphr.h"
#include "freertos/timers.h"
#include "esp_timer.h"

static SemaphoreHandle_t xSemaphore = NULL;

#define I2C_SDA 8
#define I2C_SCL 9
#define I2C_FRQ 100000
#define I2C_PORT 0

// #define STATS_BUFFER_SIZE 1024
// #define portCONFIGURE_TIMER_FOR_RUN_TIME_STATS() /* Configure a hardware timer if necessary */
// #define portGET_RUN_TIME_COUNTER_VALUE() xTaskGetTickCount()

// static char stats_buffer[STATS_BUFFER_SIZE];

// DHT11_init(GPIO_NUM_4)

max30100_config_t max30100 = {};

// Define global variables to store task execution times
int64_t get_bpm_time = 0;
int64_t dht11_time = 0;

esp_err_t i2c_master_init(i2c_port_t i2c_port)
{
    i2c_config_t conf = {};
    conf.mode = I2C_MODE_MASTER;
    conf.sda_io_num = I2C_SDA;
    conf.scl_io_num = I2C_SCL;
    conf.sda_pullup_en = GPIO_PULLUP_ENABLE;
    conf.scl_pullup_en = GPIO_PULLUP_ENABLE;
    conf.master.clk_speed = I2C_FRQ;
    i2c_param_config(i2c_port, &conf);
    return i2c_driver_install(i2c_port, I2C_MODE_MASTER, 0, 0, 0);
}

/*void get_bpm(void* param) {
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(true) {
        // Update sensor, saving to "result"
        ESP_ERROR_CHECK(max30100_update(&max30100, &result));
        if(result.pulse_detected) {

```

```

        printf("BEAT\n");
        printf("BPM: %f | SpO2: %f%\n", result.heart_bpm, result.spO2);
    }
    //Update rate: 100Hz
    vTaskDelay(1000/portTICK_PERIOD_MS);
}
}*/

void get_bpm(void* param)
{
    printf("MAX30100 Test\n");
    max30100_data_t result = {};
    while(1)
    {
        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

        printf("sensing task priority of Max30100 : %d
\n",uxTaskPriorityGet(NULL));
        if (xSemaphoreTake(xSemaphore, portMAX_DELAY) == pdTRUE)
        {
            // Update sensor, saving to "result"
            esp_err_t ret = max30100_update(&max30100, &result);
            if (ret != ESP_OK)
            {
                printf("Error updating MAX30100: %s\n", esp_err_to_name(ret));
            }
        }
        else
        {
            // Print a message when sensor data is updated
            printf("Sensor data updated successfully\n");

            // Print the pulse detection status
            printf("Pulse Detected: %s\n", result.pulse_detected ? "Yes" :
"No");

            if(result.pulse_detected)
            {
                // Print a message when a pulse is detected
                printf("BPM: %f | SpO2: %f%\n", result.heart_bpm,
result.spO2);
            }
        }
    }
}

int64_t end_time = esp_timer_get_time(); // End time in microseconds

```

```

        get_bpm_time = end_time - start_time; // Calculate processing time for
get_bpm task

        // Update rate: 100Hz
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void dht11(void *data)
{
    while(1)
    {

        int64_t start_time = esp_timer_get_time(); // Start time in
microseconds

        printf("sensing task priority of DHT11 : %d
\n", uxTaskPriorityGet(NULL));
        printf("Temperature is %d \n", DHT11_read().temperature);
        printf("Humidity is %d\n", DHT11_read().humidity);
        printf("Status code is %d\n", DHT11_read().status);

        // Give the semaphore to signal that data is ready
        xSemaphoreGive(xSemaphore);

        printf("Semaphore given by dht11 task\n");

        int64_t end_time = esp_timer_get_time(); // End time in microseconds
        dht11_time = end_time - start_time; // Calculate processing time for
dht11 task

        vTaskDelay(1000/portTICK_PERIOD_MS);
    }
}

/*void print_task_runtime_stats(void *params)
{
    while(1) {
        vTaskGetRunTimeStats(stats_buffer);
        printf("Task runtime stats:\n%s\n", stats_buffer);
        vTaskDelay(5000 / portTICK_PERIOD_MS); // Print stats every 5 seconds
    }
}*/

void app_main()
{

```

```

DHT11_init(GPIO_NUM_4);
//Init I2C_NUM_0
ESP_ERROR_CHECK(i2c_master_init(I2C_PORT));
//Init sensor at I2C_NUM_0
ESP_ERROR_CHECK(max30100_init( &max30100, I2C_PORT,
                               MAX30100_DEFAULT_OPERATING_MODE,
                               MAX30100_DEFAULT_SAMPLING_RATE,
                               MAX30100_DEFAULT_LED_PULSE_WIDTH,
                               MAX30100_DEFAULT_IR_LED_CURRENT,
                               MAX30100_DEFAULT_START_RED_LED_CURRENT,
                               MAX30100_DEFAULT_MEAN_FILTER_SIZE,
                               MAX30100_DEFAULT_PULSE_BPM_SAMPLE_SIZE,
                               true, false ));

// Create the semaphore
xSemaphore = xSemaphoreCreateBinary();
if (xSemaphore == NULL)
{
    printf("Failed to create semaphore\n");
    return;
}
//Start test task
xTaskCreate(get_bpm, "Get BPM", 2048, NULL, 1, NULL);
xTaskCreate(dht11, "dht11", 2048, NULL, 2, NULL);
//xTaskCreate(print_task_runtime_stats, "Print Task Stats", 2048, NULL, 1,
NULL);

while(1)
{
    printf("main task : %d \n", uxTaskPriorityGet(NULL));
    printf("get_bpm CPU utilization time: %lld ms \n", get_bpm_time);
    printf("dht11 CPU utilization time: %lld ms \n", dht11_time);

    vTaskDelay(1000/portTICK_PERIOD_MS);
}
}

```

OUTPUT :

```
ESP-IDF 5.2 CMD - "C:\MAJOR" x + v
sensing task priority of DHT11 : 2
MAX30100 Test
sensing task priority of Max30100 : 1
Temperature is 31
Humidity is 65
Status code is 0
Semaphore given by dht11 task
main task : 1
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 24527 ms
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 65
Status code is 0
Semaphore given by dht11 task
main task : 1
get_bpm CPU utilization time: 32681 ms
dht11 CPU utilization time: 506 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
sensing task priority of Max30100 : 1
Temperature is 31
```

when 2 cores are used

```
main task : 1
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 0 ms
sensing task priority of DHT11 : 2
MAX30100 Test
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
Temperature is 32
Humidity is 63
Status code is 0
main task : 1
get_bpm CPU utilization time: 8167 ms
```

dht11 CPU utilization time: 24748 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1194 ms

dht11 CPU utilization time: 360 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1157 ms

dht11 CPU utilization time: 24168 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1207 ms

dht11 CPU utilization time: 351 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1166 ms

dht11 CPU utilization time: 24159 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1207 ms

dht11 CPU utilization time: 349 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1166 ms

dht11 CPU utilization time: 24158 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1
get_bpm CPU utilization time: 1209 ms
dht11 CPU utilization time: 349 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 32
Humidity is 63
Status code is 0

main task : 1
get_bpm CPU utilization time: 1168 ms
dht11 CPU utilization time: 24156 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 32
Humidity is 63
Status code is 0

main task : 1
get_bpm CPU utilization time: 1209 ms
dht11 CPU utilization time: 349 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 32
Humidity is 63
Status code is 0

main task : 1
get_bpm CPU utilization time: 1168 ms
dht11 CPU utilization time: 24156 ms
sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1209 ms

dht11 CPU utilization time: 349 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1168 ms

dht11 CPU utilization time: 24156 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63

Status code is 0

main task : 1

get_bpm CPU utilization time: 1209 ms

dht11 CPU utilization time: 349 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 32

Humidity is 63
Status code is 0
main task : 1
get_bpm CPU utilization time: 1168 ms
dht11 CPU utilization time: 24156 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 32
Humidity is 63
Status code is 0
main task : 1
get_bpm CPU utilization time: 1209 ms
dht11 CPU utilization time: 349 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 32
Humidity is 63
Status code is 0

when only one core is used

sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
MAX30100 Test
sensing task priority of Max30100 : 1
main task : 1
get_bpm CPU utilization time: 0 ms

dht11 CPU utilization time: 24379 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

main task : 1

get_bpm CPU utilization time: 10731 ms

dht11 CPU utilization time: 364 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

sensing task priority of Max30100 : 1

main task : 1

get_bpm CPU utilization time: 1163 ms

dht11 CPU utilization time: 24197 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

main task : 1

get_bpm CPU utilization time: 9765 ms

dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31
Humidity is 64
Status code is 0
sensing task priority of Max30100 : 1
main task : 1
get_bpm CPU utilization time: 3058 ms
dht11 CPU utilization time: 24184 ms
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
main task : 1
get_bpm CPU utilization time: 9761 ms
dht11 CPU utilization time: 350 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
sensing task priority of Max30100 : 1
main task : 1
get_bpm CPU utilization time: 3056 ms
dht11 CPU utilization time: 24183 ms
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
main task : 1

get_bpm CPU utilization time: 9765 ms

dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

sensing task priority of Max30100 : 1

main task : 1

get_bpm CPU utilization time: 3058 ms

dht11 CPU utilization time: 24185 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

main task : 1

get_bpm CPU utilization time: 9760 ms

dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

sensing task priority of Max30100 : 1

main task : 1

get_bpm CPU utilization time: 3056 ms

dht11 CPU utilization time: 24183 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

main task : 1

get_bpm CPU utilization time: 9765 ms

dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

sensing task priority of Max30100 : 1

main task : 1

get_bpm CPU utilization time: 3058 ms

dht11 CPU utilization time: 24185 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

main task : 1

get_bpm CPU utilization time: 9760 ms

dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0
sensing task priority of Max30100 : 1
main task : 1
get_bpm CPU utilization time: 3056 ms
dht11 CPU utilization time: 24183 ms
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
main task : 1
get_bpm CPU utilization time: 9765 ms
dht11 CPU utilization time: 350 ms
sensing task priority of Max30100 : 1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
sensing task priority of Max30100 : 1
main task : 1
get_bpm CPU utilization time: 3058 ms
dht11 CPU utilization time: 24184 ms
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
Temperature is 31
Humidity is 64
Status code is 0
main task : 1
get_bpm CPU utilization time: 9761 ms
dht11 CPU utilization time: 350 ms

sensing task priority of Max30100 : 1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

Temperature is 31

Humidity is 64

Status code is 0

core output is also shown means it gives info about in which core which task is running and both task running in one core

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

MAX30100 Test

sensing task priority of Max30100 : 1

sensing core max :0

main task : 1

get_bpm CPU utilization time: 0 ms

dht11 CPU utilization time: 24455 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 12636 ms

dht11 CPU utilization time: 448 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

sensing task priority of Max30100 : 1

sensing core max :0

main task : 1

get_bpm CPU utilization time: 1230 ms

dht11 CPU utilization time: 24271 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 11672 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 6888 ms

dht11 CPU utilization time: 24262 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 10032 ms

dht11 CPU utilization time: 436 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

sensing task priority of Max30100 : 1

sensing core max :0

main task : 1

get_bpm CPU utilization time: 7219 ms

dht11 CPU utilization time: 24260 ms

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 11673 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 6897 ms

dht11 CPU utilization time: 24264 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 66

Status code is 0

main task : 1

get_bpm CPU utilization time: 10032 ms

dht11 CPU utilization time: 436 ms

sensing task priority of Max30100 : 1

sensing core max :0

Sensor data updated successfully

Pulse Detected: No

core output is also shown means it gives info about in which core which task is running and both task running in two core

MAX30100 Test

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 5584 ms

dht11 CPU utilization time: 24517 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1267 ms

dht11 CPU utilization time: 4057 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24213 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1278 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24207 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1268 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24207 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1278 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24206 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1268 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24209 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1278 ms

dht11 CPU utilization time: 442 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0

main task : 1

get_bpm CPU utilization time: 1221 ms

dht11 CPU utilization time: 24207 ms

sensing task priority of Max30100 : 1

sensing core max :1

Sensor data updated successfully

Pulse Detected: No

sensing task priority of DHT11 : 2

sensing core DHT11 : 0

Temperature is 31

Humidity is 65

Status code is 0
main task : 1
get_bpm CPU utilization time: 1268 ms
dht11 CPU utilization time: 442 ms
sensing task priority of Max30100 : 1
sensing core max :1
Sensor data updated successfully
Pulse Detected: No
sensing task priority of DHT11 : 2
sensing core DHT11 : 0
Temperature is 31
Humidity is 65
Status code is 0
main task : 1
get_bpm CPU utilization time: 1221 ms
dht11 CPU utilization time: 24210 ms

when only utilization time caluculated without considering the output

MAX30100 Test
sensing core DHT11 : 0
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 324 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 5009 ms
dht11 CPU utilization time: 105 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1040 ms

dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1040 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0
get_bpm CPU utilization time: 1040 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1
sensing core DHT11 : 0

get_bpm CPU utilization time: 1041 ms
dht11 CPU utilization time: 90 ms
sensing task priority of Max30100 : 1
sensing core max :1

when only utilization time caluculated without considering the output and running in single core

sensing core DHT11 : 0
MAX30100 Test
sensing task priority of Max30100 : 1
sensing core max :0
get_bpm CPU utilization time: 0 ms
dht11 CPU utilization time: 236 ms
sensing core DHT11 : 0
get_bpm CPU utilization time: 4125 ms
dht11 CPU utilization time: 94 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3558 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3560 ms
dht11 CPU utilization time: 89 ms
sensing task priority of Max30100 : 1
sensing core max :0
sensing core DHT11 : 0
get_bpm CPU utilization time: 3560 ms
dht11 CPU utilization time: 89 ms

[illegible]